Unit 5 Graph (Modification)

Title: Social Networking Application using Graph with user and friendship management.

problem statement:

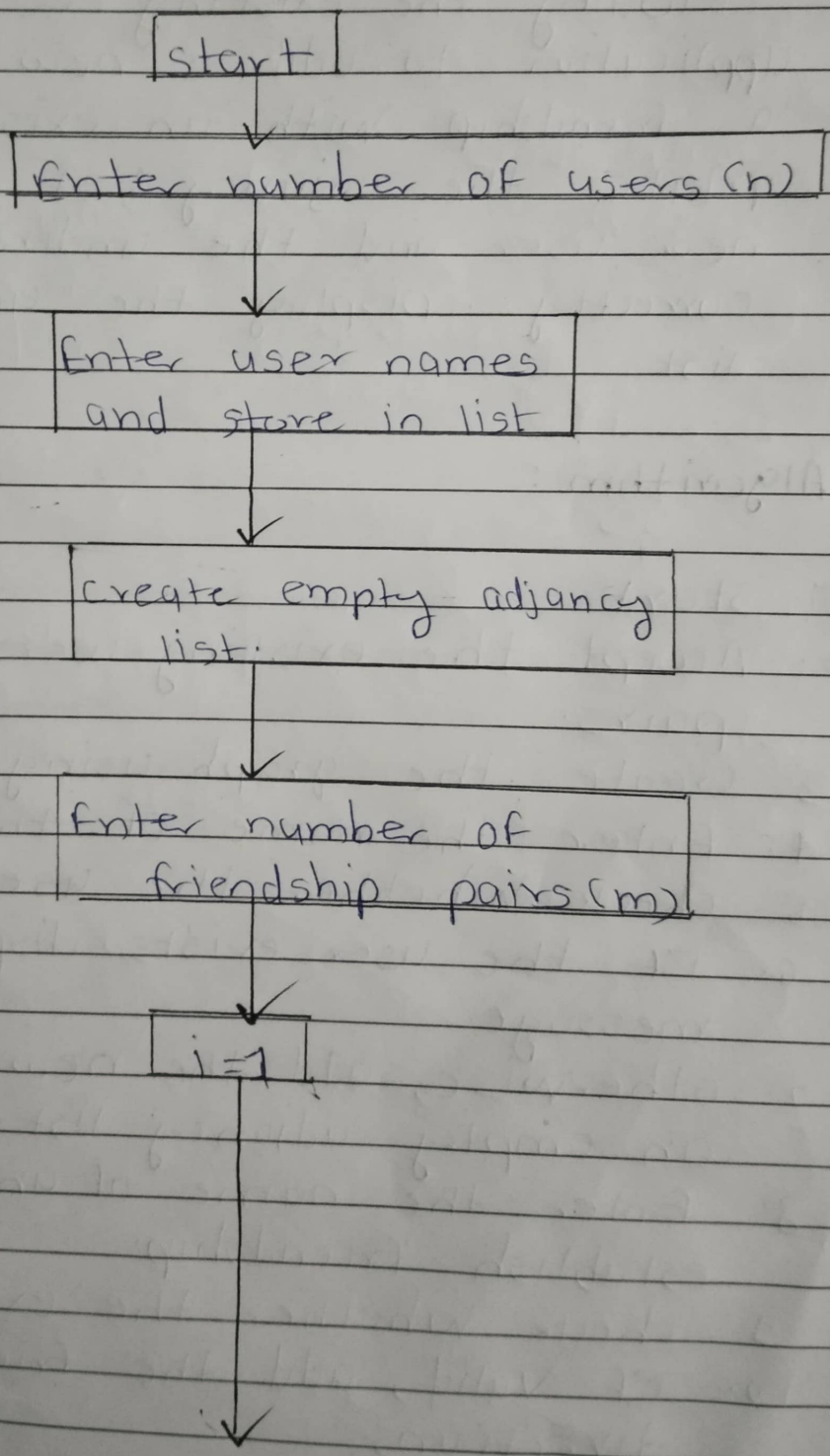
Modify the existing c++ social networking application to add a new user and establish a friendship with an existing user. update the adjancy list to represent the new user and the undirected friendship comectly. Display the updated adjacency list.

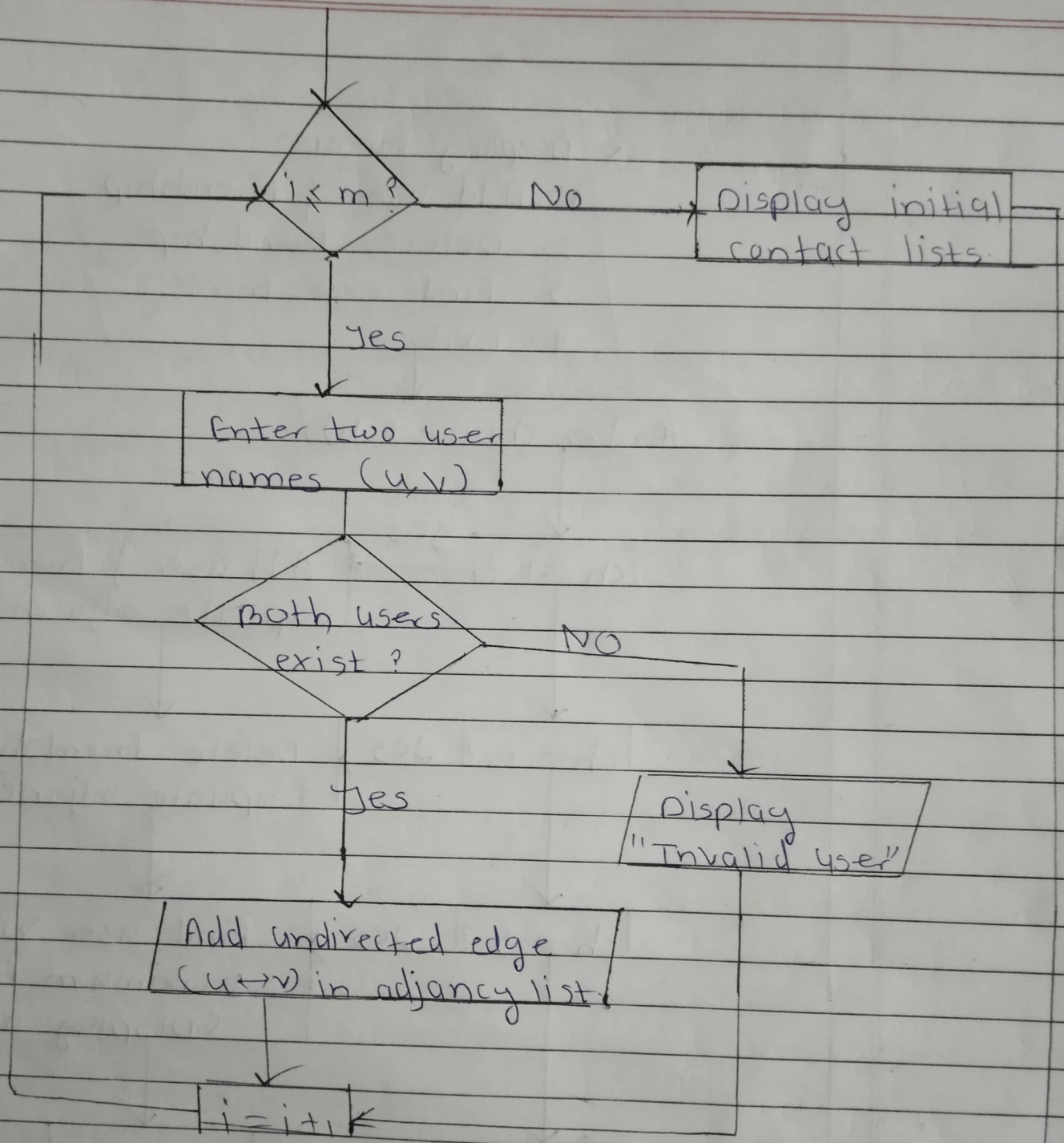
Algorithm:

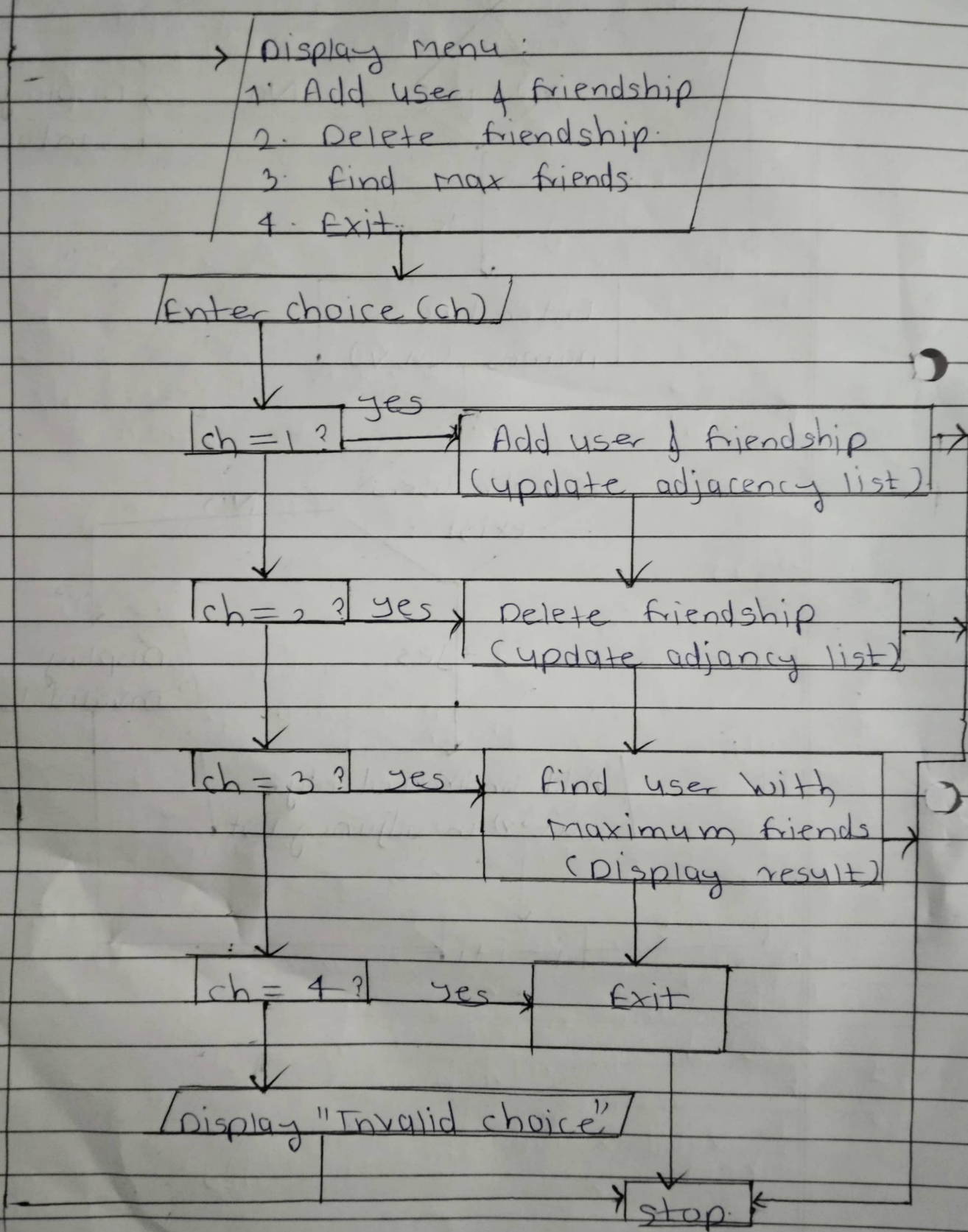
1. Start
2. Accept the existing users and friendship pairs.
3. Create the graph using an adjacency list.
4. Enter the name of the new user.
5. check whether the user alredy exists.
6. If the user exists, display an appropriate message.
7. otherwise, add the new user and create an empty adjancy list.
8. Enter the name of an existing user to establish friendship.
9. check whether the existing user is valid.
10. If valid, add the friendship in both directions.

11. Display the updated adjancy list.
12. Stop.

Flowchart:




i ≤ m ?
No
Display initial contact lists.
Yes
Enter two user names (u, v)
Both users exist ?
No
Yes
Display "Invalid user"
Add undirected edge (u ↔ v) in adjancy list.
i = i + 1


Display Menu:
1. Add user & friendship
2. Delete friendship
3. Find max friends
4. Exit
Enter choice (ch)
ch = 1 ?
yes
Add user & friendship (update adjacency list)
ch = 2 ?
yes
Delete friendship (update adjancy list)
ch = 3 ?
yes
Find user with maximum friends (Display result)
ch = 4 ?
yes
Exit
Display "Invalid choice"
stop

```
#include <iostream>
#include <vector>
#include <string>
#include <unordered_map>
using namespace std;

int main()
{
   int n, m;
   cout << "Enter number of users: ";
   cin >> n;

   if (n <= 0)
   {
      cout << "Invalid number of users.\n";
      return 0;
   }

   vector<string> users;
   vector<vector<int>> adj;
   unordered_map<string, int> index;

   cout << "Enter user names:\n";

   for (int i = 0; i < n; i++)
   {
      string name;
      cin >> name;
```

```
    if (index.count(name))
    {
        cout << "Duplicate user name.\n";
        return 0;
    }

    index[name] = users.size();
    users.push_back(name);
    adj.push_back({});
}

cout << "Enter number of friendship pairs: ";
cin >> m;

if (m < 0)
{
    cout << "Invalid number of friendships.\n";
    return 0;
}

cout << "Enter friendship pairs:\n";

for (int i = 0; i < m; i++)
{
    string u, v;
    cin >> u >> v;
```

```
    if (!index.count(u) || !index.count(v) || u == v)
    {
        cout << "Invalid friendship pair.\n";
        continue;
    }

    adj[index[u]].push_back(index[v]);
    adj[index[v]].push_back(index[u]);
}

string newUser, friendName;

cout << "Enter new user name: ";
cin >> newUser;

if (index.count(newUser))
{
    cout << "User already exists.\n";
    return 0;
}

index[newUser] = users.size();
users.push_back(newUser);
adj.push_back({});

cout << "Enter friend name: ";
cin >> friendName;
if (!index.count(friendName) || friendName == newUser)
```

```
    {
        cout << "Invalid friend name. New user added without friendship.\n";
    }
    else
    {
        int u = index[newUser];
        int v = index[friendName];

        adj[u].push_back(v);
        adj[v].push_back(u);

        cout << "User and friendship added successfully.\n";
    }
    cout << "\nUpdated Adjacency List:\n";

    for (int i = 0; i < (int)users.size(); i++)
    {
        cout << users[i] << " -> ";

        if (adj[i].empty())
            cout << "No friends";

        for (int j : adj[i])
            cout << users[j] << " ";
        cout << "\n";
    }
    return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programing code
Problems Output Debug Console Terminal Ports
powershell
PS D:\c programing code> g++ code.cpp
PS D:\c programing code> ./a.exe
Enter number of users: 5
Enter user names:
ashish
shubham
yash
siddhant
rockey
Enter number of friendship pairs: 5
Enter friendship pairs:
ashish shubham
ashish yash
shubham siddhant
yash rockey
siddhant rockey
Adjacency List:
ashish -> shubham yash
shubham -> ashish siddhant
yash -> ashish rockey
siddhant -> shubham rockey
rockey -> yash siddhant
Enter user name to check friends: ashish
Friends of ashish: shubham yash
PS D:\c programing code>
0 0 BLACKBOX Agent Open Website
Ln 128, Col 1 Spaces: 4 UTF-8 CRLF { } C++ Go Live AI Code Chat Win32
Search
ENG IN
74%
16:49
25-10-2026